

October 16, 2025

Directive Number: 3A

Group: 1

Dan Caverly, Matthew Drummond, Michael Hicks, Sean Stack

As of October 16, 2025, satisfactory progress has been made on the Group Data Project. There have been no changes to the data source or objectives of the selected models. Data cleaning has been completed, and model testing/training has begun.

The following steps have been completed:

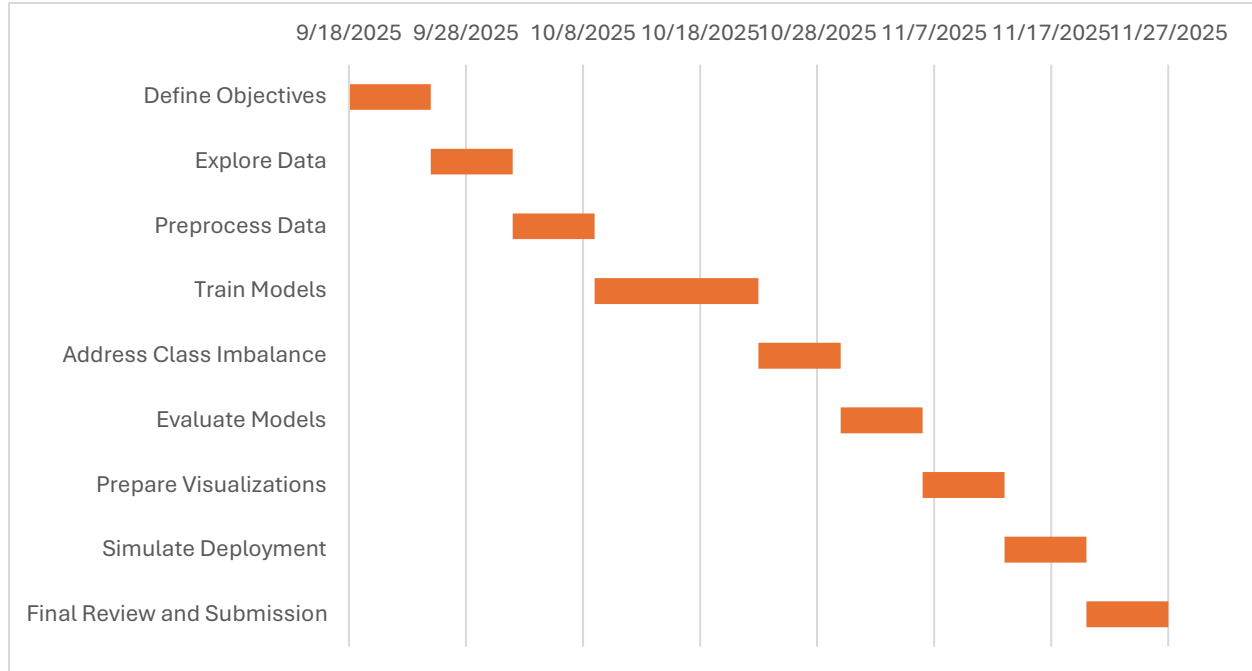
1. Data Integration
 - a. Left join train_transaction and train_identity on transactionID
 - b. drop columns with > 95% missing values
2. Outlier and Zero transaction handling
 - a. remove transactions with TransactionAmt == 0
 - b. flag top 1% transaction amounts
3. Missing Value Treatment
 - a. Numerical Features
 - i. impute per class (fraud vs legit) using median
 - ii. Create binary feature_missing indicator for columns with <20% missing
 - b. Categorical Features
 - i. If <95% missing, impute with "missing"
4. Encoding
 - a. Label encoding for high-cardinality categorical features
 - b. One-hot encoding for low-cardinality categorical features

These steps have been performed in one R script.

Current Goals:

- Build a successful model with no coding errors
 - Reveals insightful information
 - Challenges our hypothesis
- Develop strong visualizations that convey a message to the audience

Gantt Chart:



When viewing our Gantt chart, it is clear that we are right on schedule. Currently, our team is undergoing testing and training our models. As of October 16th, 2025, we are on track to finish the project on time according to the schedule. However, due to the volume of tasks to complete within the “training models” section, we expect to go slightly over the anticipated timeframe. However, it is possible that the team can exceed timetable expectations on another part of the project. In order to combat this, our team will designate 1-2 members to begin preparing visualizations. This is a step that can be completed now that data cleaning is finished. The only issue that is holding us back is knowing exactly what visualizations are necessary at this time. It is much more likely that we will know the proper approach during the “address class imbalance” phase. It is still too early to tell if we will require more time to complete this project. But at our current state, we should have no problem finishing on time.